

Problems 1, 2, and 3 require you to have read the lecture notes or watched the lecture, but should be doable.

Problem 4 is important to have seen and attempted: you will go through it during the tutorial (but it is worth giving it some thought before). Problem 5 is recommended, and while the analysis to get the final bound on C is technical and somewhat annoying, it is a good idea to try to attempt the rest.

Problem 6 is quite technical and long (especially the question marked with a $(\star)$): it is alright to skip it, or to skip that subquestion if you attempt the problem. Attempt Problem 7 if you have time: it is not necessary, but gives perspective on the use of LSH.

Warm-up

Problem 1. Give a data structure for the Nearest Neighbour problem over a d -dimensional universe using space $O(nd)$, for which QUERY runs in time $O(nd)$. (Also, show that it can maintain S dynamically, and implement INSERT and REMOVE methods running in time $O(nd)$.)

Problem 2. Give a data structure for the Nearest Neighbour problem over $\{0,1\}^d$ using space $O(2^d)$, for which QUERY runs in time $O(2^d)$ (independent of n). (Also, can maintain S dynamically, and implement INSERT and REMOVE methods running in time $O(1)$.)

Problem 3. Check your understanding: since we want very efficient lookups and are willing to accept a small probability of failure for QUERY, can we use Bloom filters for the “baby version” of LSH instead of hash tables? What fails?

Problem solving

Problem 4. $(\star)$ Prove a simplified version of Theorem 39 from the lecture notes, showing how to solve the “general” ANN from the “baby version,” at the cost of only a logarithmic factor in the ratio

$$\Delta = \frac{\max_{x,x' \in S} \text{dist}(x, x')}{\min_{x,x' \in S} \text{dist}(x, x')}$$

Note that, for the Hamming space $\{0,1\}^d$, $\Delta = O(d)$, where d is the dimension.

Problem 5. Analyse the LSH family described in the lecture notes for the Euclidean case, where a locally-sensitive hash function $h_g: \mathbb{R}^d \rightarrow \{-1,1\}$ is obtained by drawing a d -dimensional Gaussian random vector $g \sim \mathcal{N}(0_d, I_d)$ (all coordinates are independent $\mathcal{N}(0,1)$ normal random variables) and setting

$$h_g: x \in \mathbb{R}^d \rightarrow \text{sign}\left(\sum_{i=1}^d g_i x_i\right)$$

We will make the (restrictive) assumption that all data points and query points have unit norm: $\|x\|_2 = 1$. Show that, for every $r > 0, C > 1$, this defines an (r, C, p, q) -LSH family with p, q such that $\rho \leq 1/C$. [Note: this is called the *SimHash* scheme.]

Problem 6. (★) For the set $[d] = \{1, 2, \dots, d\}$, let the universe $\mathcal{X}$ be the set of all 2^d subsets of $[d]$, along with the *Jaccard distance*:

$$\text{dist}(A, B) = 1 - \frac{|A \cap B|}{|A \cup B|}, \quad A, B \in \mathcal{X}$$

Consider the following hash family $\mathcal{H}$: for every permutation $\pi: [d] \rightarrow [d]$, define $h_\pi: \mathcal{X} \rightarrow [d]$ by setting

$$h_\pi(A) = \min_{a \in A} \pi(a)$$

and $\mathcal{H} = \{h_\pi\}_\pi$.

- a) (★) Verify that the Jaccard distance is a metric on $\mathcal{X}$. What is its range?
- b) What is the size of $\mathcal{H}$?
- c) Show that, for every $r \in (0, 1]$ and $C > 1$, $\mathcal{H}$ is an (r, C, p, q) -LSH family for $p = 1 - r$ and $q = 1 - Cr$. What is its sensitivity parameter ρ ?

Advanced

Problem 7. Give a data structure for the Nearest Neighbour problem over the Euclidean space $(\mathbb{R}^d, \ell_2)$ based on kd-trees. Analyse the space complexity of the data structure and its query time.